

CONCEPT PROPOSAL

BARCELONA RACE

A 21st-Century Team-Building Adventure

Turn a traditional Race Around the World into a live, location-aware game.
Teams walk Barcelona, unlock clues on arrival, submit evidence, and compete on a server clock.

1. AT A GLANCE

This document is a working concept for a friend-led team-building afternoon in Barcelona. It is written so you can decide whether to run the first event, what to build, and what to leave open.

The afternoon

Four teams of four. One official game phone per team. Ten to twelve checkpoints on a walkable loop. About 90–150 minutes of racing, plus briefing and a finish drink.

The game

Reach a place → GPS unlocks the clue → solve the challenge → submit a photo or answer → AI or the organiser judges → next checkpoint.

How you win

Lowest adjusted race time wins. Points are a tie-break and a live-leaderboard flavour, not a second championship.

What is modern about it

A mobile web app instead of paper clues. Server-side time. Geofenced unlocks. AI for routine photo judging. A live organiser dashboard and map.

What this is not

Not a claim that no GPS hunt exists in Barcelona. Local operators and apps already do versions of this. The case for building is control, custom clues, and a reusable engine — or it is a learning project. See §8.

Ask of you

Pick the area and date, freeze team names, and walk the loop once. Software follows the route, not the other way around.

RECOMMENDED FIRST MOVE

Do not start by writing the whole application. Freeze 10 checkpoints and their rules, walk them with a phone, then build the smallest app that can run that loop on four real devices.

2. THE IDEA

The result is part scavenger hunt, part Amazing Race, part city exploration — delivered through a mobile web app rather than paper clues and clipboards. Teams move between checkpoints, unlock clues only when they are physically there, solve mixed challenges, and compete against a clock the phones cannot fake.

Core principles

- **Simple.** One designated game phone per team. A second phone is allowed only for the uses listed in the rules.
- **Physical.** Reaching the place matters. GPS can unlock a checkpoint; it does not complete it.
- **Competitive.** Time, penalties and submissions live on the server.
- **Flexible.** Challenges can be photo, trivia, observation, puzzle, search, QR or interaction.
- **Modern, not automatic.** AI judges routine photos. Ambiguous cases go to a human in one click.
- **Fair in a real city.** Tight geofences in the Gothic Quarter will fail. Design for urban GPS, crowds and closed doors.

3. HOW THE GAME WORKS

Teams. Pilot size is 4 × 4. The engine should not hard-code four teams. Each team gets a name, colour and PIN.

Game phone. One phone is the official device. It supplies GPS, receives clues, captures official photos and shows game status. Keep that tab open; iOS in particular will not track reliably in the background.

Second phone. Allowed for maps, translation, and research challenges. Not allowed as a second GPS source, a second camera for official evidence, or a way to split the team and farm checkpoints. Recommendation for the first event: teams stay together.

Race clock. The server is the official source of time. Phones only display it.

Checkpoint flow. Reach location → checkpoint unlocks → receive clue → solve challenge → submit evidence → AI or human verification → result → next checkpoint. A skip-with-penalty option stops a team from dying on one puzzle.

EXAMPLE PLAYER SCREEN — CHECKPOINT 4

Locked until the team enters the geofence.

Clue: “Find the place where the city watches the world go by.”

Challenge: Photograph a public clock showing a time later than 15:00.

On accept, the next checkpoint unlocks. On reject, a two-minute penalty is applied and they may resubmit. After the challenge time cap they may skip for a larger penalty.

4. GPS AND CHECKPOINT ACTIVATION

The browser requests location with explicit permission. The game receives latitude, longitude and an estimated accuracy. A checkpoint is a coordinate plus an activation radius.

Recommended first-event radii: **60–80 metres** in open squares and the Eixample; **80–120 metres** in the Gothic Quarter and other urban canyons. A 15-metre “stand on this cobble” fence will produce disputes, not drama.

BARCELONA REALITY

Smartphone GNSS under open sky is often around 5 metres. Between tall streets it is commonly 15–50 metres, with occasional wrong-side-of-the-street fixes. Barcelona’s dense old town is a known difficult environment. GPS must be one proof layer, never the only one. Combine geofence + photo, QR or answer. Walk every fence with the same class of phone the teams will use, at the time of day you will race.

Record on every GPS event

- Team ID, checkpoint ID, latitude, longitude, accuracy, timestamp, and whether the fence accepted the fix.
- If accuracy is worse than the fence radius, refuse the unlock and ask the team to wait for a better signal — or offer an organiser override.
- Show the player a simple GPS status: good / poor / unavailable. Do not make them guess why the clue is still locked.

5. PHOTO CHALLENGES AND AI JUDGING

A vision model evaluates a submitted photograph against a written specification. That avoids building a custom computer-vision system per challenge. The model returns correct / incorrect / ambiguous, a confidence score, a short reason, and a penalty recommendation.

Example specification: “Find a red scooter parked beside a blue door.” Must contain a scooter, predominantly red, a blue door, both in the same frame. Reject if colour or object is ambiguous, or if one required element is missing.

GREEN — auto-accept
clearly correct

AMBER — organiser decides
ambiguous / low confidence

RED — auto-reject
clearly wrong

AI never has absolute authority. Amber cases land on a one-click organiser queue.

Compress photos on the device before upload. Store the file against team, checkpoint, server timestamp and GPS fix at submit time. Strip or ignore EXIF location after judging so a photo archive is not also a movement archive.

6. SUGGESTED CHALLENGE TYPES

Type	Example	Verification
Photo	Photograph a red door with a lion's head.	AI vision, amber to organiser
Trivia	Find the year a building was inaugurated.	Exact answer / rule
Search	Use the second phone to confirm a historical fact visible on site.	Answer, requires presence
Puzzle	Solve a puzzle presented at the location.	Answer or organiser
Interaction	Find someone wearing a team shirt and take a photo.	AI + human; consent first
Observation	How many windows on the second floor of this façade?	Answer; must be on site
Navigation	Infer the next location from a riddle.	GPS arrival
QR	Find a hidden QR that unlocks the next stage.	QR + GPS

Research and observation challenges fail if the answer is the first Google result from a café three streets away. Write them so the team has to be there. Interaction challenges must never require photographing unwilling strangers.

7. SCORING, PENALTIES AND TIMING

The backend is authoritative for race time, checkpoint timestamps, submissions and penalties. That prevents teams from manipulating a phone clock.

Recommended first-event model

WHO WINS

Primary ranking: lowest adjusted race time.

Adjusted time = (finish clock – start clock) + penalty seconds + skip seconds.

Correct challenges award +100 points. Points are the public leaderboard flavour and the tie-break if two teams finish within 30 seconds of each other on adjusted time. Do not run two championships at once.

- Wrong submission: +120 seconds. The team may resubmit.
- Skip / failed challenge after the time cap: +300 seconds, then proceed.
- Challenge time cap: 8 minutes from clue unlock. Clock keeps running either way.
- A wrong attempt costs time only when it is wrong. Sitting on a puzzle also costs time, because the race clock never stops.

Timestamp every important action: GPS reached → clue unlocked → evidence submitted → AI result → accepted or rejected → next checkpoint. Those events are the dispute log.

8. BUILD VERSUS BUY

Barcelona already has GPS treasure hunts and Amazing Race-style team building. Operators run hotspot hunts with live scoreboards. Platforms such as GooseChase, Scavify, HuntHopper, PlayTours and Let's Roam already offer photo missions, GPS check-ins, QR, live leaderboards, and in some cases AI photo review. This idea is not empty territory.

Build a custom app only if at least two of the following are true:

- You want full control of scoring, penalties and a server clock that no SaaS template quite matches.

- You want Barcelona-specific clues and structured AI criteria you can edit without a vendor.
- You want a reusable engine for later cities, not a one-afternoon rental.
- You accept this as a learning project and will walk the route before writing code.

If the goal is simply “four teams, one Saturday, no engineering,” configure an existing platform and spend the effort on the route. The rest of this document assumes you do want the custom engine.

9. ANTI-CHEATING AND FAIRNESS

- Official game phone is the only source of GPS and official photo evidence.
- Server time is the official race clock.
- GPS accuracy is checked before a tight fence can open.
- Checkpoint completion requires GPS plus a challenge, not GPS alone.
- Photos are timestamped and bound to a team and checkpoint.
- Every game event is logged for disputes.
- Ambiguous AI decisions go to the organiser instead of auto-penalising.
- Second-phone rules are written on one page and briefed before start.
- Teams stay together on the first event. Splitting turns the race into a logistics contest and a safety problem.

10. LIVE ORGANISER DASHBOARD

The organiser has a separate admin view. Click a team to see last GPS position, accuracy, current checkpoint, submitted evidence, AI result, penalties and event history. A live map with team markers is useful once the core loop works; it is not required for the first code drop.

Team	Checkpoint	Status	Race time	Penalties
Red	7 / 10	Moving	01:14:32	2 min
Blue	6 / 10	Submitting	01:18:04	0 min
Green	7 / 10	Moving	01:21:17	4 min
Yellow	5 / 10	Stuck — offer skip	01:25:09	0 min

The dashboard also needs two unglamorous buttons: **manual unlock** (dead GPS, courtyard, phone on the blink) and **override AI**. Those two controls will save the afternoon more often than a prettier map.

11. TECHNICAL ARCHITECTURE

Separate the public game interface from live game state. GitHub holds source and deployment. Live race state lives in a real backend. GitHub Pages is fine for a static brochure; it cannot run this game.

Layer	Recommendation	Note
Frontend	Next.js / React PWA	Test on iPhone Safari, not only Android Chrome.
Hosting	Vercel, Fly.io or equivalent	Needs server routes and environment secrets.
Backend	Serverless API or small Node/Python service	Authoritative clock and scoring.
Database	Postgres via Supabase or similar	Teams, events, submissions, penalties.
Photos	Object storage	Compress on device; short retention.
AI	Vision-capable API	Structured criteria in, verdict out.

Layer	Recommendation	Note
Maps	Mapbox, Google Maps or OSM	Add after the loop works on four phones.
Realtime	Supabase Realtime or short polling	Leaderboard and admin map.
Development	AI-assisted coding against the repo	Build in the phases in §13.

Budget three things people forget: vision-API cost per photo, mobile-data size of uploads, and battery drain from continuous geolocation. A player rule — official phone on charge pack, screen awake at checkpoints — is part of the product.

12. PROJECT STRUCTURE AND CONFIGURATION

Challenges are configuration, not code. The organiser should be able to change locations, clues, time limits, points and AI criteria without rewriting the application.

```
barcelona-race/  
  app/team/      player PWA  
  app/admin/     organiser dashboard  
  app/api/       scoring, GPS, uploads, AI  
  components/    TeamDashboard, Countdown, GPSStatus,  
                 CameraCapture, ChallengeCard  
  game/          checkpoints, scoring, geofencing, rules  
  challenges/    barcelona.json  
  README.md
```

Example checkpoint

```
{  
  "checkpoint": 4,  
  "name": "Gothic Quarter",  
  "latitude": 41.3839,  
  "longitude": 2.1761,  
  "radiusMeters": 90,  
  "clue": "Find the creature watching over the narrow street.",  
  "challengeType": "photo",  
  "instruction": "Photograph the gargoyle.",  
  "aiCriteria": ["stone gargoyle", "mounted on building", "outdoors"],  
  "rewardPoints": 100,  
  "wrongPenaltySeconds": 120,  
  "skipPenaltySeconds": 300,  
  "timeLimitSeconds": 480  
}
```

Radius is 90 m here on purpose. A 60 m fence in that street will fight the architecture.

13. BUILD SEQUENCE

Build in small, testable increments. Do not ask any coding agent — human or otherwise — for the entire application in one prompt.

Phase	What ships	Done when
1 Engine	Teams, PINs, checkpoints, clues, scoring, penalties, server clock.	A scripted race can be completed without GPS or photos.
2 Player PWA	Login, current checkpoint, countdown, clue, submit answer.	Four phones can play a desk race.
3 GPS	Permission, live fix, accuracy gate, geofence unlock.	A real fence opens and refuses a poor fix.

Phase	What ships	Done when
4 Photos	Capture, compress, upload, store, bind to team/checkpoint.	A photo survives a flaky connection.
5 AI judging	Criteria in, green / amber / red out, organiser queue.	A known-good and known-bad photo grade correctly.
6 Admin	Leaderboard, review queue, manual unlock, override.	You can unstick a team without touching the database.
7 Hardening	Poor GPS, duplicates, two teams in one fence, dead phone, offline upload.	A walk-through in Barcelona does not need a laptop rescue.
8 Polish	Live map, richer challenge types, QR, analytics, visual design.	Only after phase 7.

Useful small tasks: “Add a 90-metre geofence to checkpoint 3.” “Add a two-minute penalty on reject.” “Create the amber review queue.” “Write tests for unlock and penalty calculation.”

14. BARCELONA CHALLENGE DESIGN

For the first event, target 10–12 checkpoints over a compact area — Gothic Quarter plus one adjacent district is enough. Each stop should feel different so the day is not “walk somewhere and photograph an object.”

- 3 photo / AI vision challenges
- 2 observation challenges
- 2 research / trivia challenges that require being on site
- 1 QR or hidden-object challenge
- 1 puzzle
- 1 human interaction challenge with consent built in
- 1–2 navigation / riddle challenges

Operational checklist before you lock the JSON

- Walk the full loop at the planned time of day. Record GPS accuracy at each fence.
- Check pedestrian access, one-way alleys, shade, stairs, and whether a mixed-fitness group can do it.
- Check opening hours and ticket lines. Do not put a challenge inside Sagrada Família or Park Güell unless you have tickets.
- Have a rain plan and a heat plan. Summer asphalt is part of the course.
- Note toilets, water, and a safe finish point with seats.
- Write a one-page safety brief: stay together, watch traffic, watch bags, call the organiser if someone is hurt or lost.
- Stagger starts by three to five minutes, or give two mirrored opening routes, so teams do not form a conga line to the same gargoyle.
- Language: clues in the language the group actually speaks. Decorative Catalan is fine; mandatory Catalan on a timed puzzle is not, unless that is the point and you brief it.

15. EXAMPLE RACE EXPERIENCE

Time	What happens
12:40	Briefing. PINs, second-phone rules, safety, “keep the game tab open.” Power banks out.
13:00	Staggered start. All teams receive clue one.
13:11	Team Red enters the fence. Clue unlocks.

Time	What happens
13:15	Red submits a photo. AI returns correct at 94% confidence. +100 points, no penalty.
13:18	Blue photo rejected. +2 minutes. They resubmit.
13:42	Organiser sees Red and Green close on the map. Blue is on the puzzle. Yellow is offered a skip.
14:30	First team finishes. Others continue or hit a hard course close.
14:45	Adjusted times published. Photos on a private recap screen, not on social media unless consented.

16. PRIVACY AND DATA

This is likely a corporate afternoon in Spain. Live location, photos, timestamps and possibly photo metadata are personal data. Treat consent and retention as part of the product, not a later legal appendix.

- **Lawful basis.** Informed, voluntary consent for a one-off game is the cleanest path. Brief it before PINs are issued.
- **Minimise.** Official game phone only for GPS and official photos. Do not track personal phones.
- **Purpose limit.** Location exists to unlock checkpoints and to unstick teams. It is not a post-event movement profile.
- **Photos of people.** Participant faces need consent if you keep or share them. Strangers are not props. No social posts of identifiable people without a separate yes.
- **Metadata.** Do not keep EXIF GPS on stored images after judging.
- **Retention.** Delete location trails and submissions 7–30 days after the event unless a dispute is open. Write the number down.
- **Processors.** Hosting, object storage and the vision API need a processing agreement and an EU-aware region if you can choose one.
- **Access.** Organiser dashboard is authenticated. Teams see only their own evidence plus the public leaderboard.

INTERACTION CHALLENGES

“Find someone in a team shirt” means a teammate or a pre-briefed host, not a tourist on the Rambla. If a challenge needs a willing extra, plant that person.

17. RECOMMENDED MVP

Ship only what is required to run one real afternoon on four phones:

- 4+ teams with PINs
- 10 configured checkpoints
- GPS geofencing with an accuracy gate
- Clue unlocking and an 8-minute challenge cap
- Photo capture, compress, upload
- AI photo verification with amber organiser queue
- 2-minute wrong-answer penalty and 5-minute skip
- Server-side race clock and adjusted-time ranking
- Live leaderboard
- Organiser review, manual unlock and AI override
- Degraded mode: queued upload, last-known map position, organiser unlock if the phone dies

Once that works on four real phones around Barcelona, add the live map, QR, richer types, analytics and polish.

18. DECISIONS BEFORE DEVELOPMENT

These are still open. Agree them on paper before writing code.

Decision	Recommendation for event one
Area and maximum walking distance	Gothic Quarter + one adjacent district. Under ~4 km.
Number of checkpoints	10. Twelve only if the walk test is easy.
Total race duration	90–150 minutes plus briefing and finish.
Second phone	Maps, translation, on-site research. Not official GPS or photos.
May teams split?	No.
Scoring	Lowest adjusted time. Points as tie-break.
Must every stop be a photo?	No. Mix the types in \$6.
How automatic is AI?	Green/red automatic. Amber always human.
Live map for organiser?	Yes as soon as phase 6 is stable. Not in the first code drop.
Dead phone / no GPS / no signal	Organiser unlock + queued upload + spare power bank.
Privacy retention	Delete trails and photos within 30 days.
Start method	Stagger teams 3–5 minutes apart.
Course close	A hard end time so the finish drink still happens.

19. BOTTOM LINE

The concept is technically realistic. The strongest version is a mobile web game with a real backend: GPS decides when a checkpoint may unlock, the challenge is the gameplay, photographs are evidence, AI handles routine visual judging, and an organiser dashboard keeps a human in charge.

What turns an ordinary scavenger hunt into a modern race is the combination of GPS + physical discovery + AI photo verification + server-side timing + live standing — plus the unromantic pieces: skip rules, manual overrides, a walked route, and a privacy limit on how long you keep anyone's afternoon.

NEXT STEP

Define the first 10 Barcelona checkpoints and their challenge rules. Walk the loop. Use that JSON as test data while the MVP is built through phases 1–7. Software follows the course.

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